



CS & IT ENGINEERING



Algorithms

Dynamic Programming

Lecture No.- 07



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

mcm
sos

Topics to be Covered



Topic

Topic

Topic

Multi-Stage Graph



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe



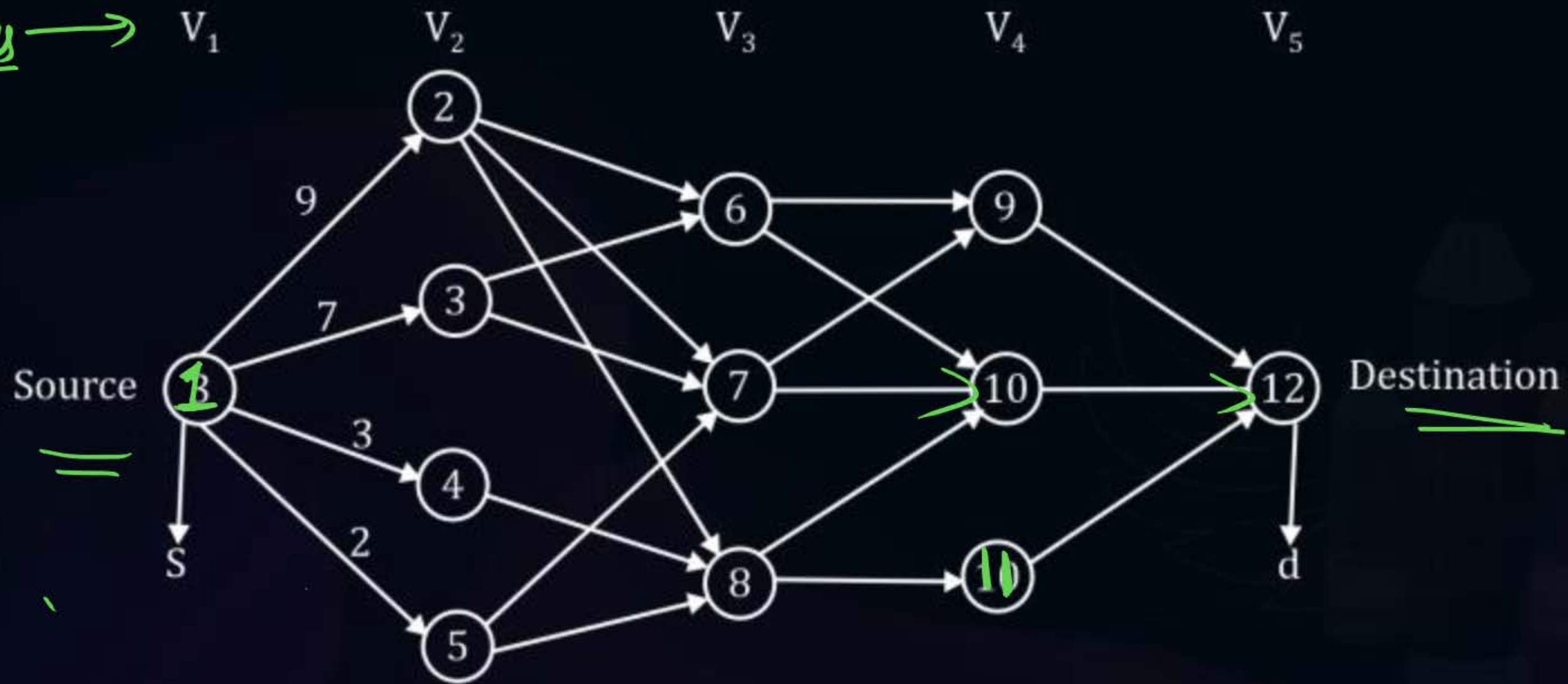
Topic : Dynamic Programming :(DP)

Eg.3.

$$V_i \rightarrow V_{i+1}$$

l-stages

Stages $\rightarrow$





Topic : Dynamic Programming :(DP)

Multi- Stage graph:

Given: $G(V,E)$

$$\underline{|V| = n}$$

$$\underline{|E| = e}$$

$$V_i \longrightarrow V_{i+1}$$

There are l stages



The vertex in any stage i can only be connected to stage $i + 1$.

$$V_i \rightarrow V_{i+1}$$



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Stage -1 and stage l (last stage) both will have only 1 vertex each.

Total 'n' vertex:

1 2 3 4 5 6

0

Remaining $(n-2)$
vertices will be in
remaining $(l-2)$ stage.

0

1

1

Source

Destination



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	1	2	3	4	5		12
1							
2							
3							
4							
5							
.							
.							
.							
12							

→ Cost adjacency Matrix

$C[i, j]$ → edge cost
e.g. $i \rightarrow j$

$C[i, j]$



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Logic:

$V \cdot V \cdot imp$

Let cost (i, j) represent cost of the path form vertex 'j' present in stage 'i' to each the destination vertex 'd'.

↓ ↓

Cost (i, j)

$C[i, j]$



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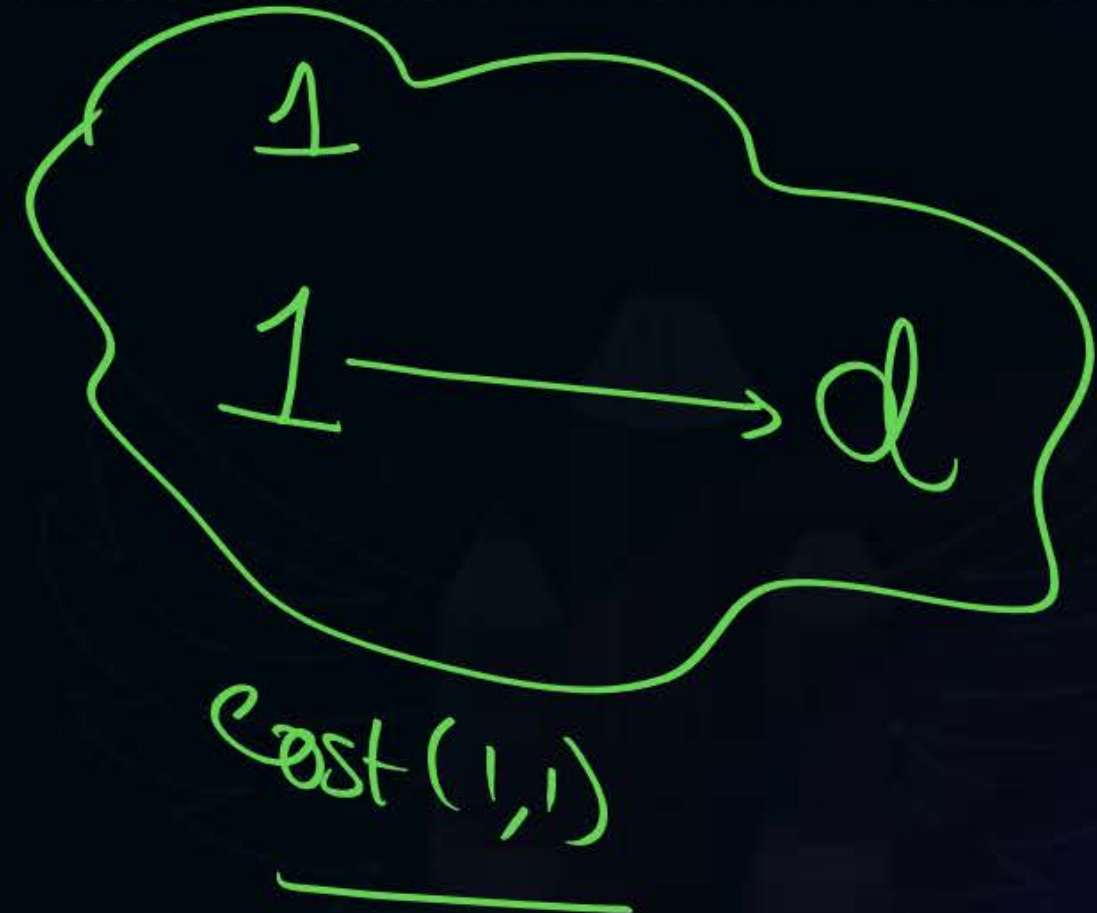
Overall problem: Cost (1,1)

Cost of the path from a vertex 1 present in stage 1 to reach the destination vertex 'd'.

Cost (i, j)

vertex 'j' present in stage 'i'

'i' $\Rightarrow$ Stage number





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$v_i \rightarrow$ Stage

$$\underline{\text{Cost}}(1,1) = \min$$

$$\boxed{K \in V_2}$$

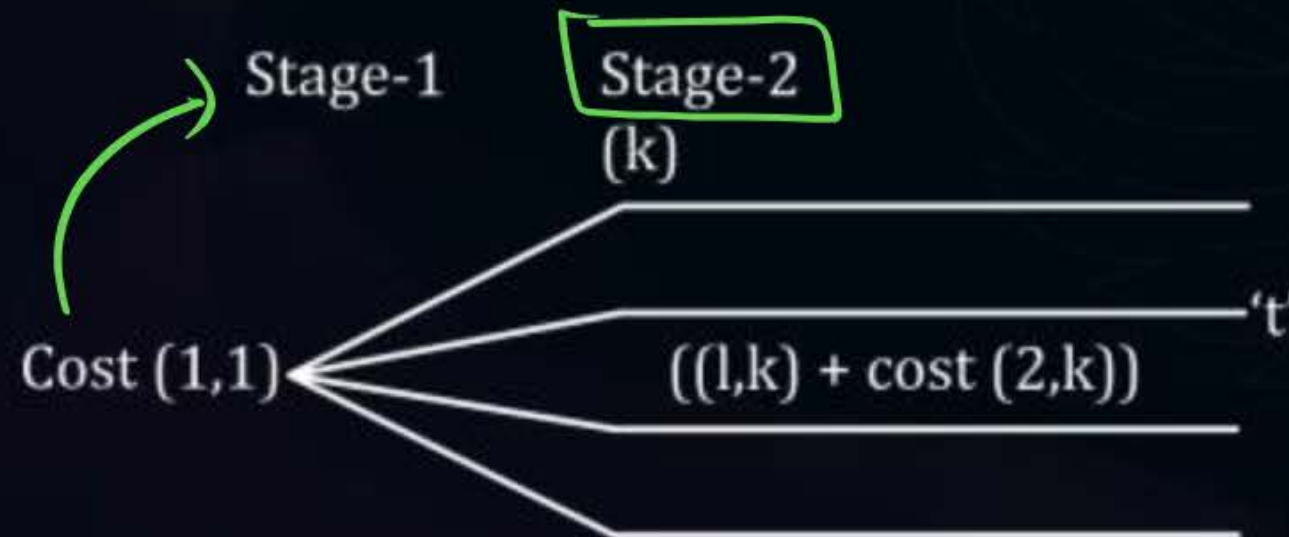
Stage-2

and

$$\langle 1, k \rangle \in E$$

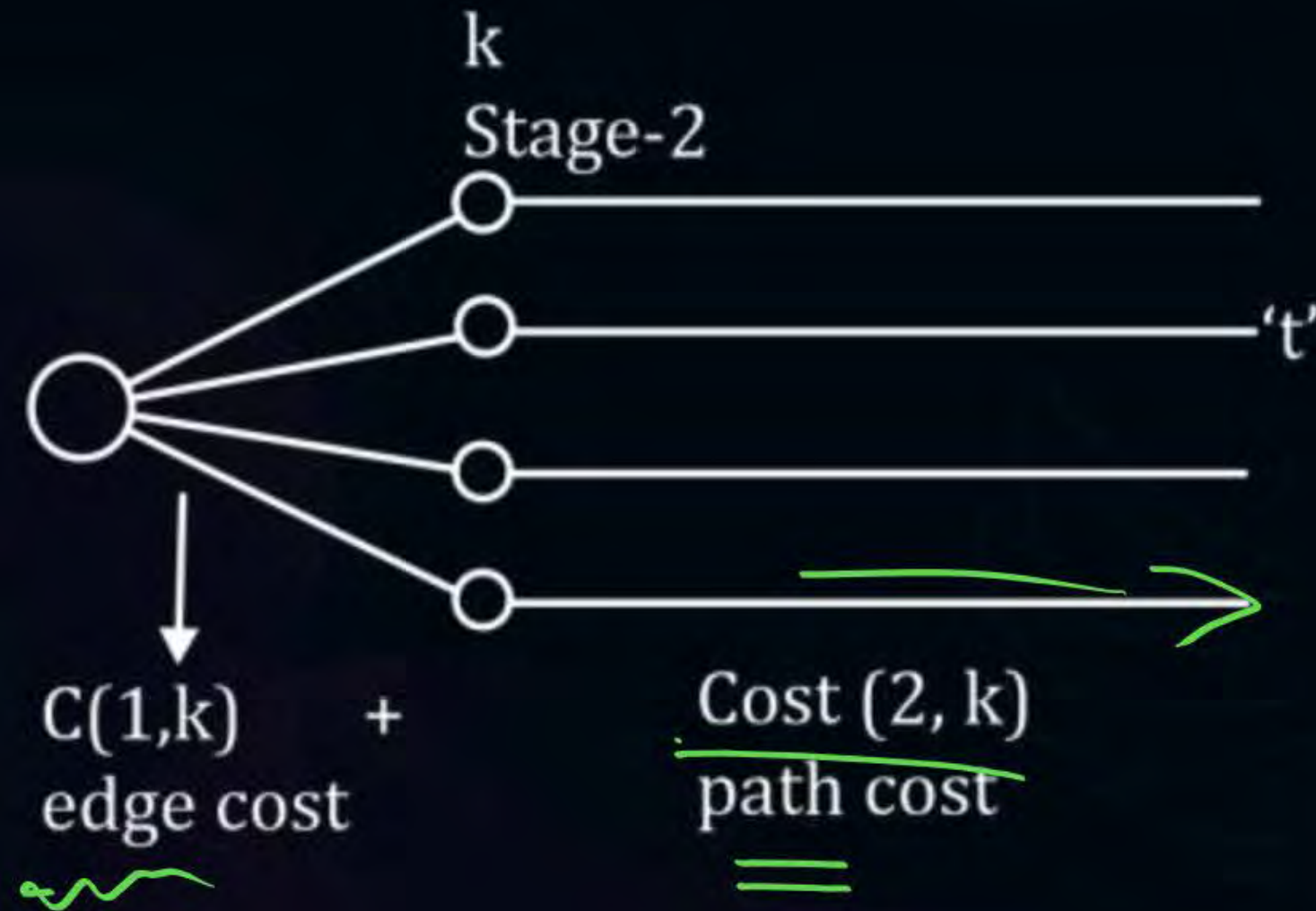
$$\left\{ \underline{C(1, k)} + \underline{\text{cost}(2, k)} \right\}$$

Subproblem





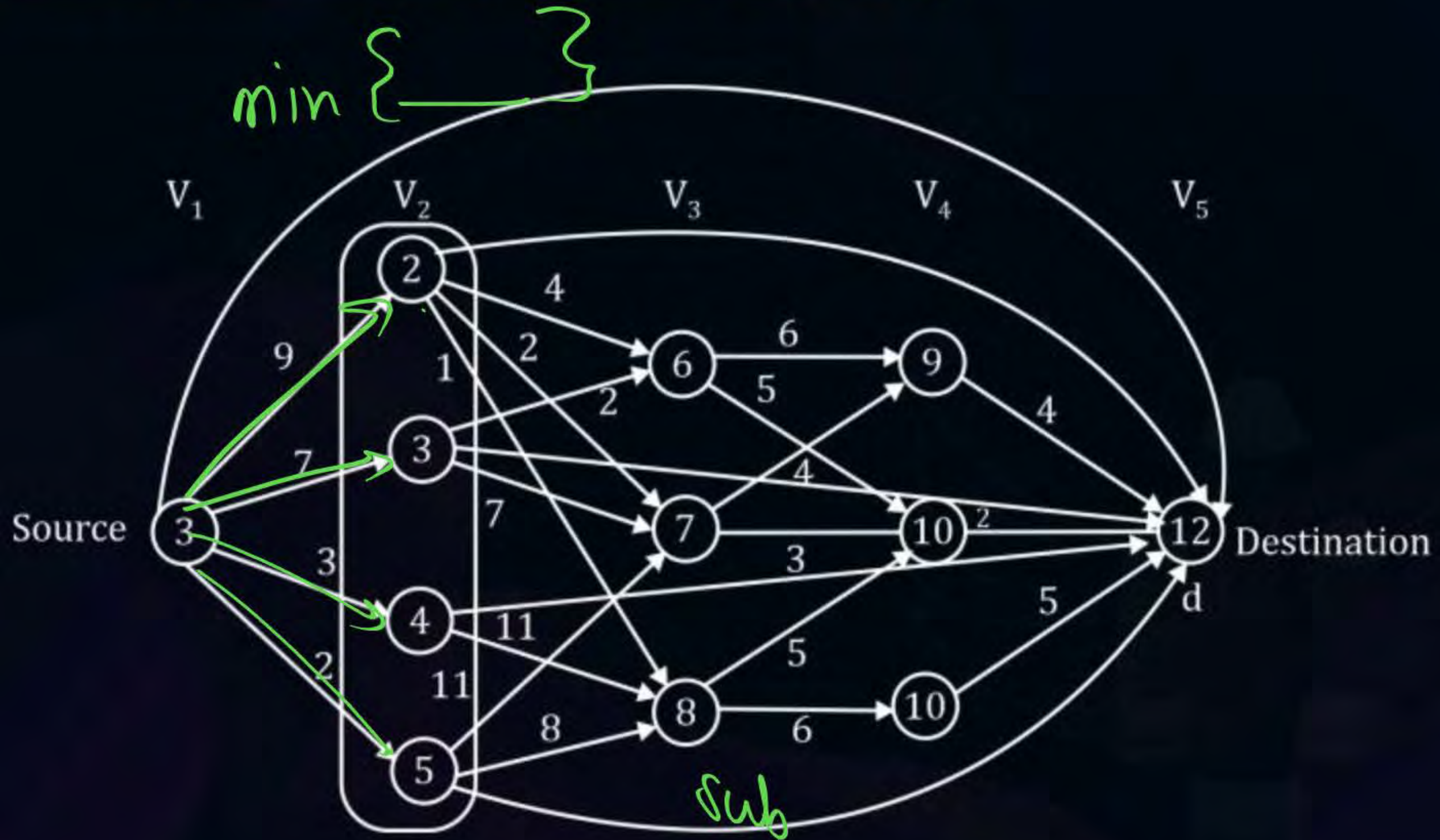
Topic : Dynamic Programming :(DP)





Topic : Dynamic Programming :(DP)

Eg. 4.

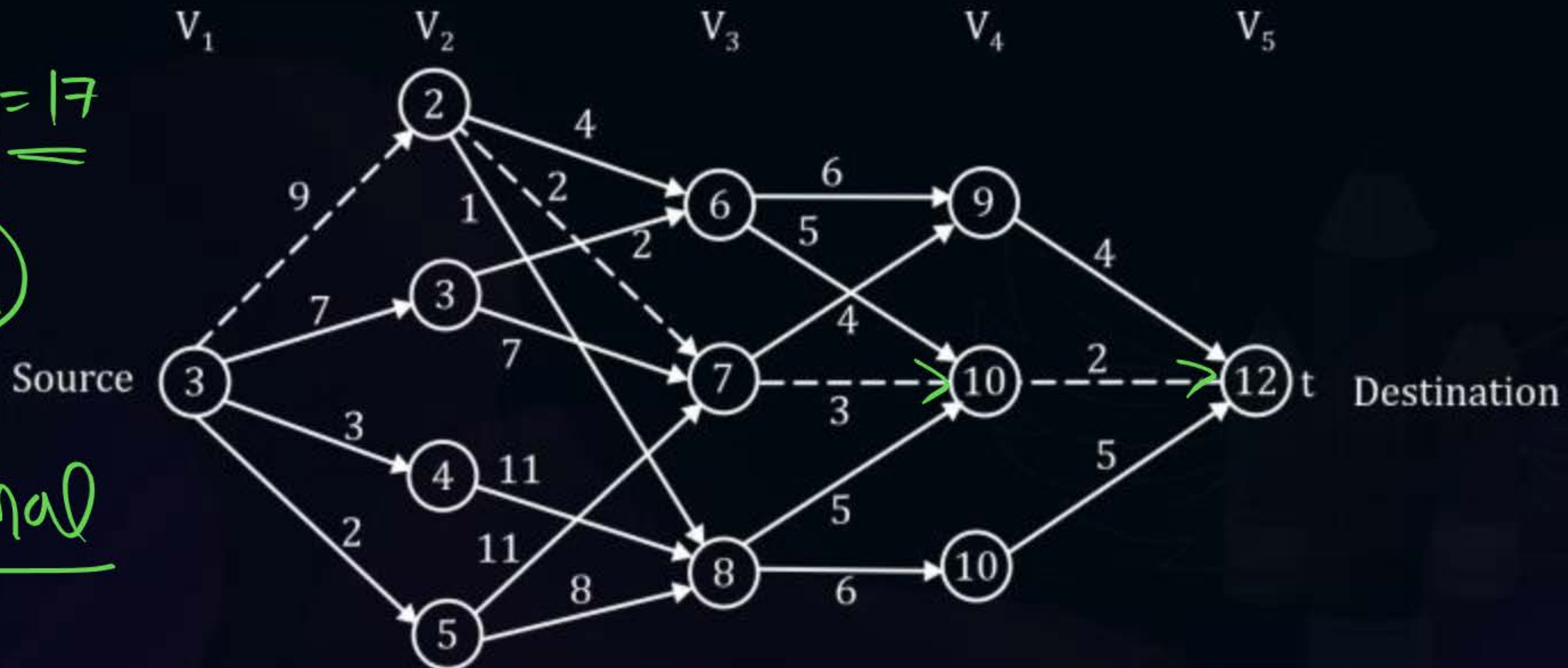




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7. Multi-Stage Graph:

Greedy = 17
DP = 16
Optimal





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Optimization Problem

Objective function

What criteria is to minimized/ maximized

(SPSP)

Minimize the cost of the path from source → Destination



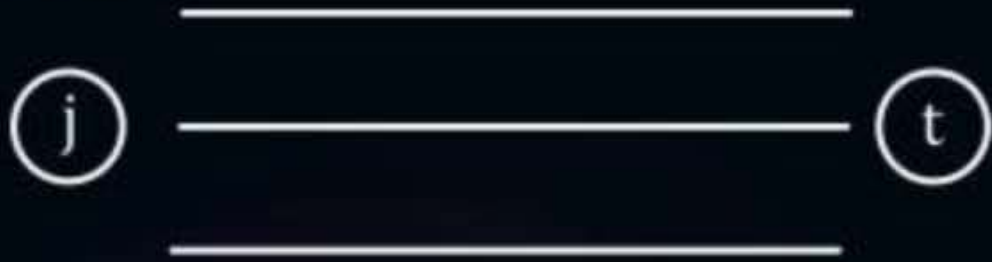
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1. $C(i,j)$: cost of edge from i^{th} vertex to j^{th} vertex
 $i \rightarrow j$
2. $\text{Cost}(i,j)$ = cost of path from vertex 'j' present in stage i to reach the destination 't'.



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Generalized Recurrence for multi-stage graph problem. 'i'



$$\text{cost}(i, j) = \min \{ c(j, k) + \text{cost}(i+1, k) \}$$

Stage number

Vertex and
 $(j, k) \in E$

$k \in V_{i+1}$

Base-Condition:

2nd last stage $(l-1)$

$$\text{cost}(l-1, j) = c(j, t)$$

$l-1$ l





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Recurrence:

$$\text{Cost}(i, j) = \min \{c(j, k) + \text{cost}(i+1, k)\} \dots (1)$$

$$K \in V_{i+1} \text{ and } (j, k) \in E$$

$l = \text{total } \del{stages} \text{ Stages}$

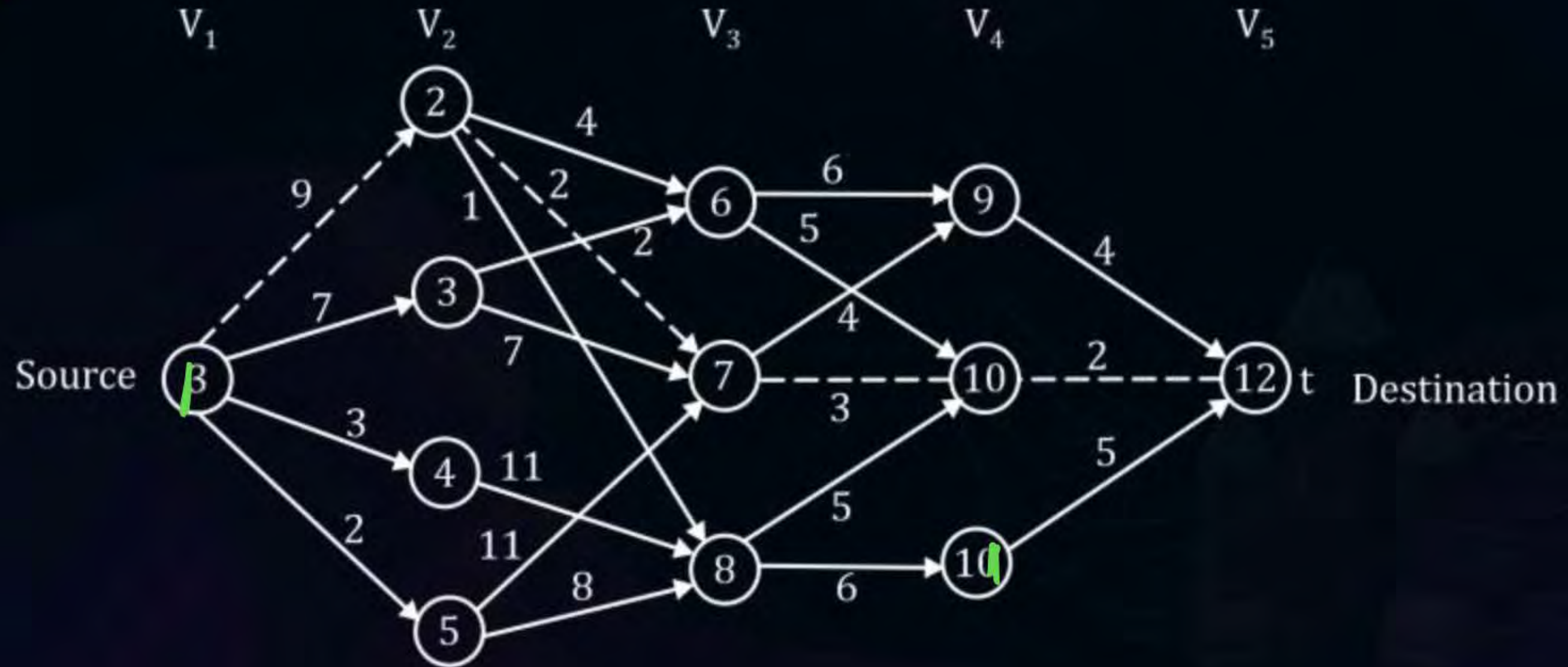
$$\text{Cost}(l-1, j) = c(j, t) \dots (2)$$

$D(i, j) = 'k'$ that minimizes equation.



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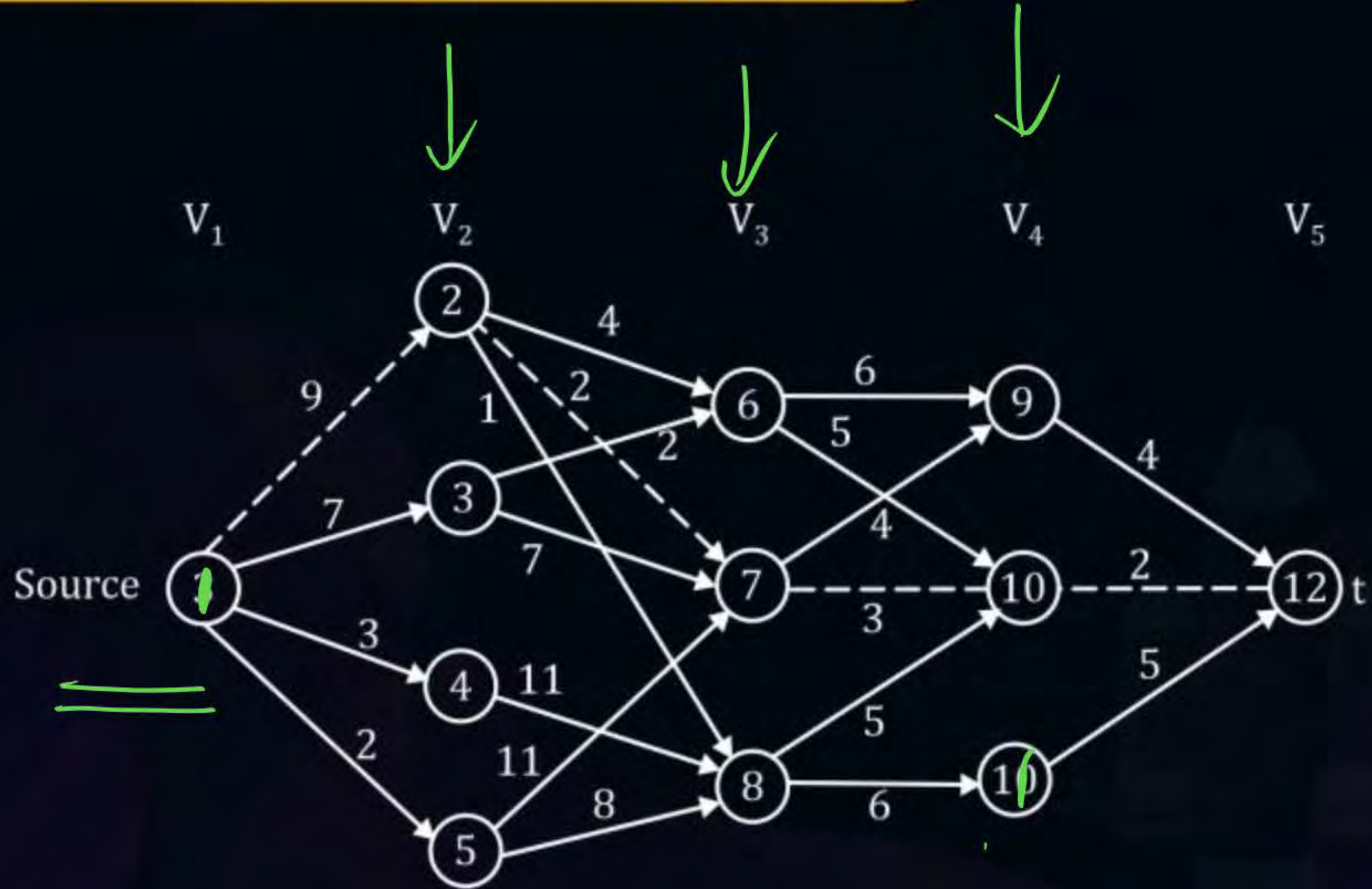
E.g. 1.





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E.g. 2.





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Stage-1

$$\underline{\text{Cost (1, 1)}} = \min \left\{ \begin{array}{l} K = 2 : (1, 2) + \underline{\text{cost (2, 2)}} \\ K = 3 : (1, 3) + \underline{\text{cost (2, 3)}} \\ K = 4 : (1, 4) + \underline{\text{cost (2, 4)}} \\ K = 5 : (1, 5) + \underline{\text{cost (2, 5)}} \end{array} \right.$$

Stage

1



2



3



4



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Stage-4

(2nd last stage)

$K = 9, 10, 11$

→ $D(4, 9) = t$ ✓

$\text{Cost}(4, 9) = c(9, t) = 4$

→ $D(4, 10) = t$ ✓

$\text{Cost}(4, 10) = c(10, t) = 2$

→ $D(4, 11) = t$ ✓

$\text{Cost}(4, 11) = c(11, t) = 5$

Stage-1



Stage-2



Stage-3



Stage-4





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Stage-3:

$J = 6, 7, 8$

$\text{Cost}(3, j)$

$$\text{Cost}(3, 6) = \text{Min} \left\{ \begin{array}{l} K=9: C(6, 9) + \text{Cost}(4, 9) \\ K=10: C(6, 10) + \text{Cost}(4, 10) \end{array} \right\}$$

$$= \text{Min} \left\{ \begin{array}{l} K=9: 6 + 4 \\ K=10: 5 + 2 \end{array} \right\} \text{ Prev Stage result}$$

$$\text{D}(3, 6) = 10$$

$$= \text{Min} \left\{ \begin{array}{l} K=9: 10 \\ K=10: 7 \end{array} \right\} \Rightarrow 7$$



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$$\text{Cost}(3,7) = \left\{ \begin{array}{l} \underline{K=9}: \quad C(7,9) + \text{Cost}(4,9) \\ \underline{K=10}: \quad C(7,10) + \text{Cost}(4,10) \end{array} \right\}$$

$$\underline{\underline{D(3,7)=10}} \quad \left\{ \begin{array}{l} K=9: \quad \underline{4+4} \\ K=10: \quad \underline{3+2} \end{array} \right\}$$

7 $\rightsquigarrow$ t

$$= \text{Min} \left\{ \begin{array}{l} K=9: \quad \underline{8} \\ K=10: \quad \underline{5} \end{array} \right\} \Rightarrow \underline{5}$$



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Cost (3,8)=

$$\text{Min} \left\{ \begin{array}{l} K = 10: C(8, 10) + \text{Cost}(4, 10) \\ K = 11: c(8, 10) + \text{Cost}(4, 11) \end{array} \right\}$$

$$\text{Min} \left\{ \begin{array}{l} K = 10: 5 + 2 \\ K = 11: 6 + 5 \end{array} \right\}$$

$$\text{Min} \left\{ \begin{array}{l} K = 10: 7 \\ K = 11: 11 \end{array} \right\} = 7$$

$$D(3, 8) = 10$$



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Stage-2:

$$J = 2, 3, 4, 5$$

$$\text{cost}(2,2) = \underline{\text{Min}} \left\{ \begin{array}{l} K=6: C(2,6) + \text{Cost}(3,6) \\ K=7: C(2,7) + \text{Cost}(3,7) \\ K=8: C(2,8) + \text{Cost}(3,8) \end{array} \right\}$$

$$= \text{Min} \left\{ \begin{array}{l} K=6: 4 + 7 \\ K=7: 2 + 5 \\ K=8: 1 + 7 \end{array} \right\}$$

$$\underline{\underline{D(2,2)=7}}$$

$$= \underline{\underline{\text{Min}}} \left\{ \begin{array}{l} K=6: 11 \\ K=7: 7 \\ K=8: 8 \end{array} \right\} = 7$$



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$$\text{Cost}(2,3) = \text{Min} \left\{ \begin{array}{l} \underline{K=6} : C(3,6) + \text{Cost}(\underline{3,6}) \\ \underline{K=7} : C(3,7) + \text{cost}(\underline{3,7}) \end{array} \right\}$$

$$\underline{D(2,3) = 6} \quad \text{Min} \left\{ \begin{array}{l} \underline{K=6} : 2 + \underline{7} \\ \underline{K=7} : 7 + \underline{5} \end{array} \right\} = \text{Min}(9, 12) = \underline{9}$$

$$\underline{D(2,4) = 8}$$

$$\text{Cost}(2,4) = \text{Min} \left\{ \underline{K=8} : \underline{11} + 7 \right\} = \underline{18}$$

$$\cancel{D(2,5)=8} \quad D(2,5)=8$$

$$\text{Cost}(2,5) = \text{Min} \left\{ \begin{array}{l} K=7 : 11 + 5 \\ \underline{K=8} : 8 + 7 \end{array} \right\} \quad \text{Min}(16, 15) = \underline{15} \quad \checkmark$$



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Stage-1

$$\text{Cost}(1,1) = \text{Min} \left\{ \begin{array}{l} K=2: C(1,2) + \text{Cost}(2,2) \\ K=3: C(1,3) + \text{Cost}(2,3) \\ K=4: C(1,4) + \text{Cost}(2,4) \\ K=5: C(1,5) + \text{Cost}(2,5) \end{array} \right\}$$

$$\underline{DP(1,1)} = ?$$

$$= \text{Min} \left\{ \begin{array}{l} 2: 9 + 7 \\ 3: 7 + 9 \\ 4: 3 + 18 \\ 5: 2 + 15 \end{array} \right\} =$$



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$$= \text{Min} \left\{ \begin{array}{cc} K & \downarrow \\ 2: & \underline{16} \\ 3: & \underline{16} \\ 4: & 21 \\ 5: & 17 \end{array} \right\} = \underline{\underline{16}}$$



$$D(1, 1) = \underline{\underline{\{2 \text{ or } 3\}}}$$



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#Q. What is the path that gives minimum cost?

$[1 \rightarrow 3 \rightarrow 6 \rightarrow 10 \rightarrow 12]$

path:- $D(1,1) = 2$

$D(2,2) = 7$

$D(3,7) = 10$

$D(4,10) = 12$

$[1 \rightarrow 2 \rightarrow 7 \rightarrow 10 \rightarrow 12]$

$D(1,1) = 3$

$D(2,3) = 6$

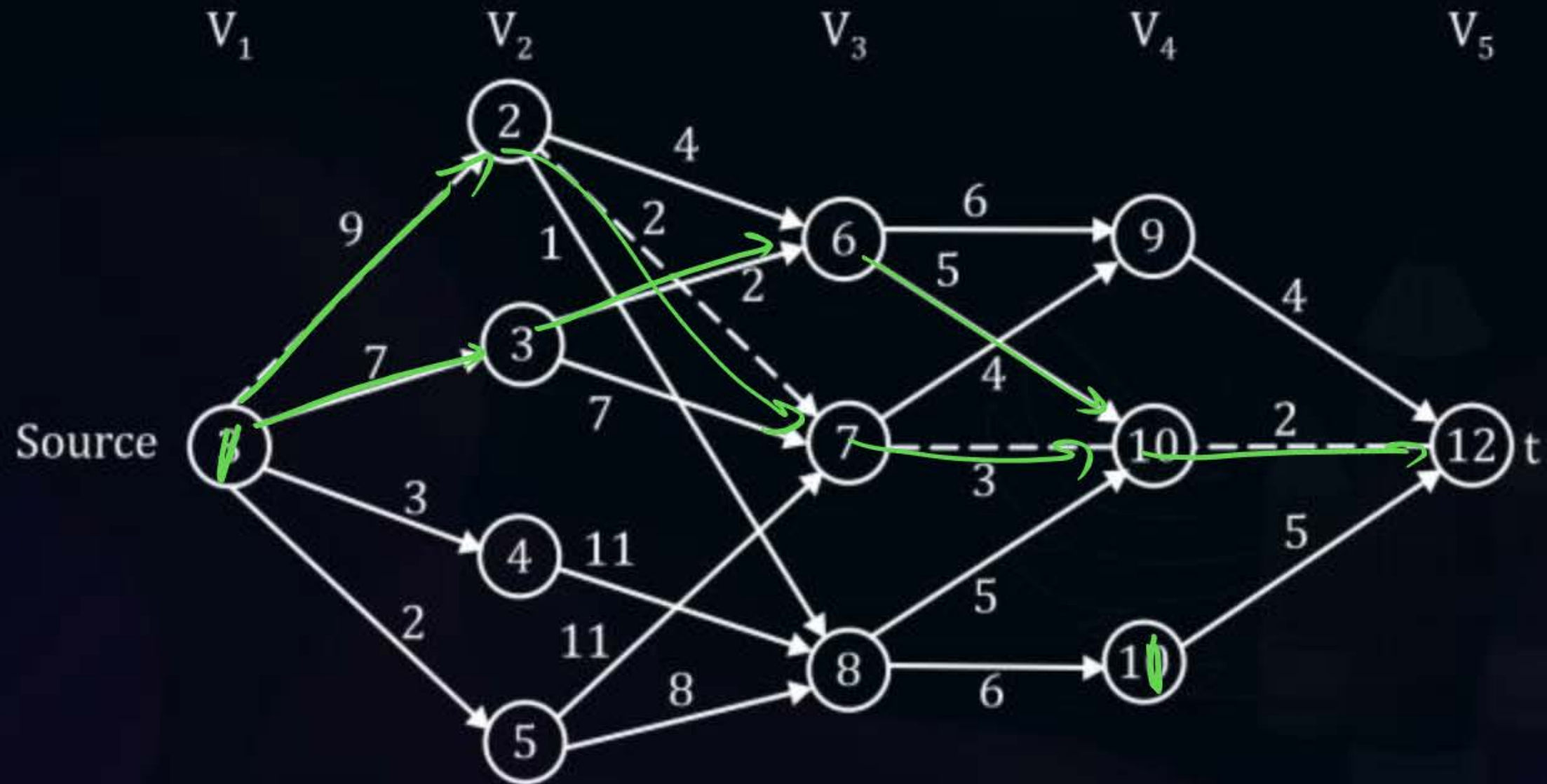
$D(3,6) = 10$

$D(4,10) = 12$



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E.g. 2.





2 mins Summary



Topic

Topic

Topic

Topic

Topic

Multi-stage graph



THANK - YOU